Ubiquitous Serial Bus

BACK IN THE EARLY 90S WHEN A LOT OF MAJOR TECHNOLOGY FIRMS HAD REALLY BOUGHT INTO THE PC ECOSYSTEM, they were faced with a major issue, that of picking the right interfaces. Every hardware manufacturer wanted users to adopt their own interface standard. And you were tied down to a small subset of all the hardware that you could purchase simply because the interface wasn't compatible.

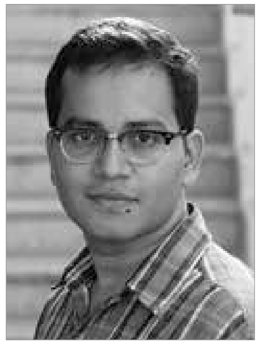
While you might not realise it when buying a cheap cable for 25 bucks from your local computer walla, there is a lot of money to be made from standards and certification. Folks buying a new PC back then could either go for Apple's Macintosh lineup of PCs or get an IBM desktop PC. Yes, the Apple vs PC debate is ancient and has actually been going on since the 80s. IBM PCs were a little more open in terms of standards and allowed for multiple vendors to build hardware accessories for the PC. The external accessories could interface using the serial port, parallel port, PS/2 port, AT-Keyboard port, SCSI ports, firewire, etc. and motherboards could simply not incorporate all these ports. Moreover, most of the interfaces could not daisy chain multiple devices so you had to decide which hardware had to be connected and which had to be chucked away. A solution had to be found.

The humble USB port started out with a very simple objective in mind, it wanted to make it fundamentally easier to connect external devices to PCs. It was small so multiple USB ports could be squeezed into the rear I/O of motherboards, it used serial communication so you could get much higher bandwidth and also daisy chain devices. Again, a lot more devices could be connected to the PC. Life became much simpler since the introduction of the USB port. Hardware vendors could rejoice since consumers could buy more devices without having to worry about lack of a compatible port to interface it with, and consumers could rejoice since shoddy hardware manufacturers could no longer lock them down to specific interfaces. Although, that didn't stop certain brands from continuing the silly practice.

Over the years, the USB port has evolved to incorporate faster speeds and stronger physical interfaces. Which is all well and good but has the USB Type-C port evolved a little too fast? The design standard was published in 2014 and within 3-4 years, the port became ubiquitous. You can find USB Type-C ports on smartphones, laptops, PCs, mirrorless cameras, TVs, consoles, water flossers, monitors, projectors, and the list keeps going on. And the USB standard has evolved quite a bit since 2014 as well. While the physical 24-pin USB Type-C interface has remained the same, the underlying data and power standards have been revised multiple times over. And therein lies the problem. With too many revisions in such a short time, the USB Type-C interface is so fragmented that it's not funny any more.

First there was the nomenclature revision wherein the logos and standards were renamed for no good reason. First, USB 3.0 went on to be called USB 3.1, then it became USB 3.2 Gen1. Aside from that, you've got USB 3.2 Gen2x1 and USB 3.2 Gen2x2. These are all the data standards. Throw in the Power Delivery spec and you've got even more names and logos which do little to help the average consumer identify which standard to go for. The Wi-Fi 802.11 standard went through a similarly rapid evolution but they finally resolved most of the confusion by switching to single digit numbers to differentiate one standard from another. There's Wi-Fi 5, followed by Wi-Fi 6, followed by Wi-Fi 7 ... wait ... scratch that, we've got to squeeze 6E in there somewhere. Let's not hope there will be a 6F, G, H, ...

I know I'm ranting about something silly but I hope you see the irony of it all. What started out as a process to make things simpler inevitably devolves into a complicated mess. At this point, one might stop calling USB as the Universal Serial Bus and simply call it Ubiquitous Serial Bus. Because it's all around you in every darned device but it sure ain't Universal by any means.



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